Lab Task 6

Task 1

Observation:

The chatbot successfully handled four test cases. For an exact question, it returned the correct answer with its source. For a similar question, it displayed possible matches with similarity scores. For empty input, it requested the user to enter a question. For unsupported input, it displayed the help desk message.

The results show that the TF-IDF retrieval system can identify relevant FAQ questions and handle empty or unsupported queries.

Task 2

Limitation:

The sentence splitter uses simple punctuation-based rules. It may incorrectly split abbreviations such as “Dr.” because it treats periods as sentence boundaries. It is a basic teaching splitter and may not work perfectly for all texts.

Task 3: Retrieval-Based Document Chatbot

Objective

The objective was to build a document retrieval system using TF-IDF and Sentence Embeddings, compare different chunking strategies, and develop an interactive chatbot.

Methodology

- Expanded the dataset to 6 documents, including conflicting information.
- Tested three chunking strategies: 40-word windows with 0 overlap, 40-word windows with 10-word overlap, and sentence-based chunking.
- Compared TF-IDF with Sentence Transformers (`all-MiniLM-L6-v2`).
- Evaluated retrieval using Hit@3 and MRR@3.
- Saved retrieval results in `retrieval_log.csv`.

Results

The selected configuration was **0-overlap chunking with Sentence Embeddings**, based on development performance.

Metric	Development	Test
Hit@3	1.00	0.75
MRR@3	1.00	0.625

The selected strategy produced 20 chunks. The maximum token length was 52, which was within the model's 256-token limit.

Chatbot Testing

The interactive chatbot was tested with:

- An assignment deadline question: relevant evidence was retrieved.
- A cafeteria prices question: the chatbot rejected it because reliable evidence was not found.
- A library renewal question: conflicting information from two documents was retrieved.
- An empty query: the chatbot displayed an appropriate message.

A threshold of **0.653** was calibrated using a small development sample to help reject unsupported queries.

Limitations

The retrieval system can return conflicting or irrelevant passages. The rejection threshold was calibrated on a small sample, so it may not work equally well for every query. Sentence-based chunking performed poorly in the held-out test.

Conclusion

Task 3 successfully implemented a retrieval-based document chatbot. The system compares retrieval strategies, displays source documents and offsets, handles empty and unsupported queries, and allows users to confirm whether they want to view retrieved evidence.